"Betting Against Beta"

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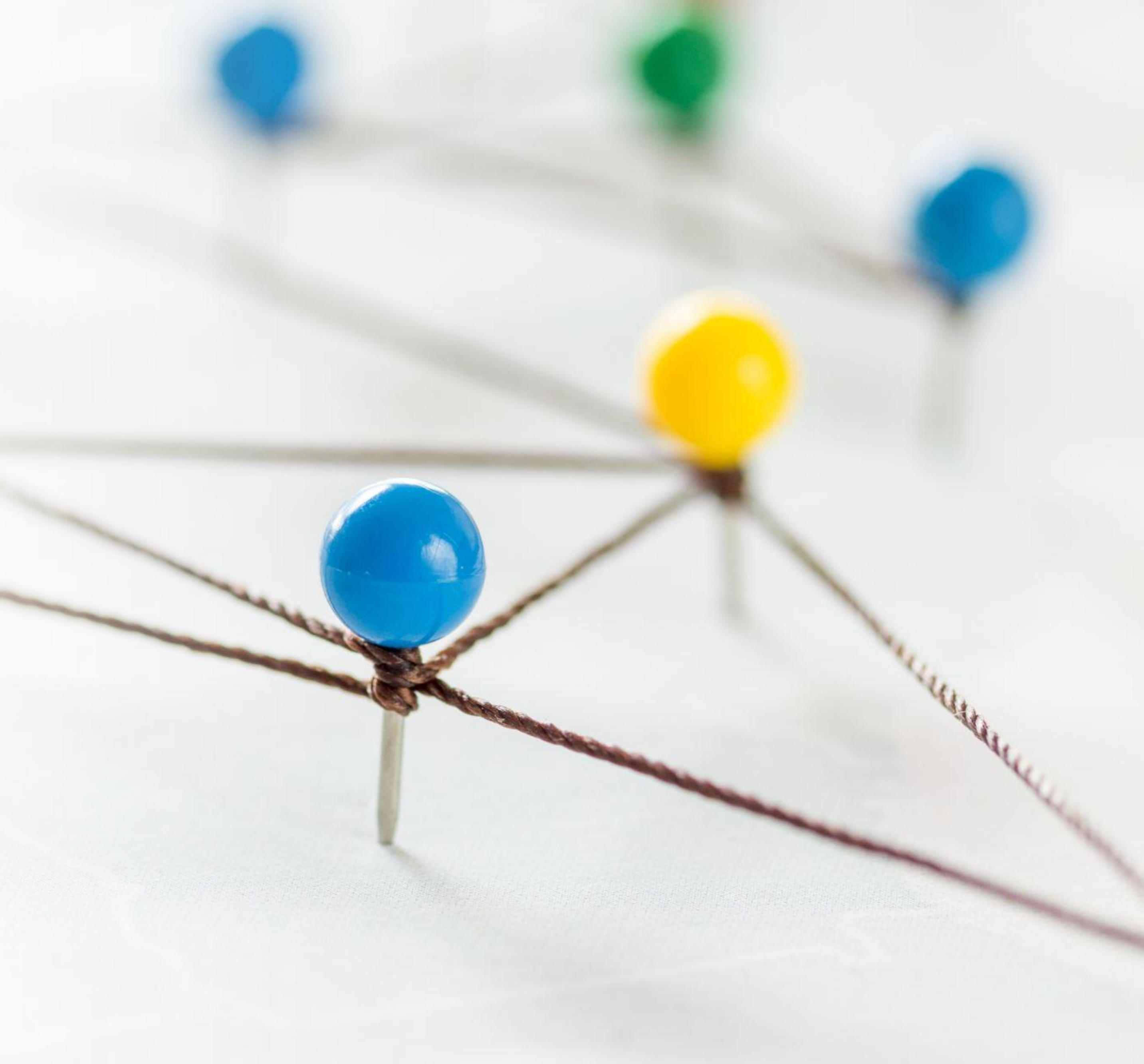


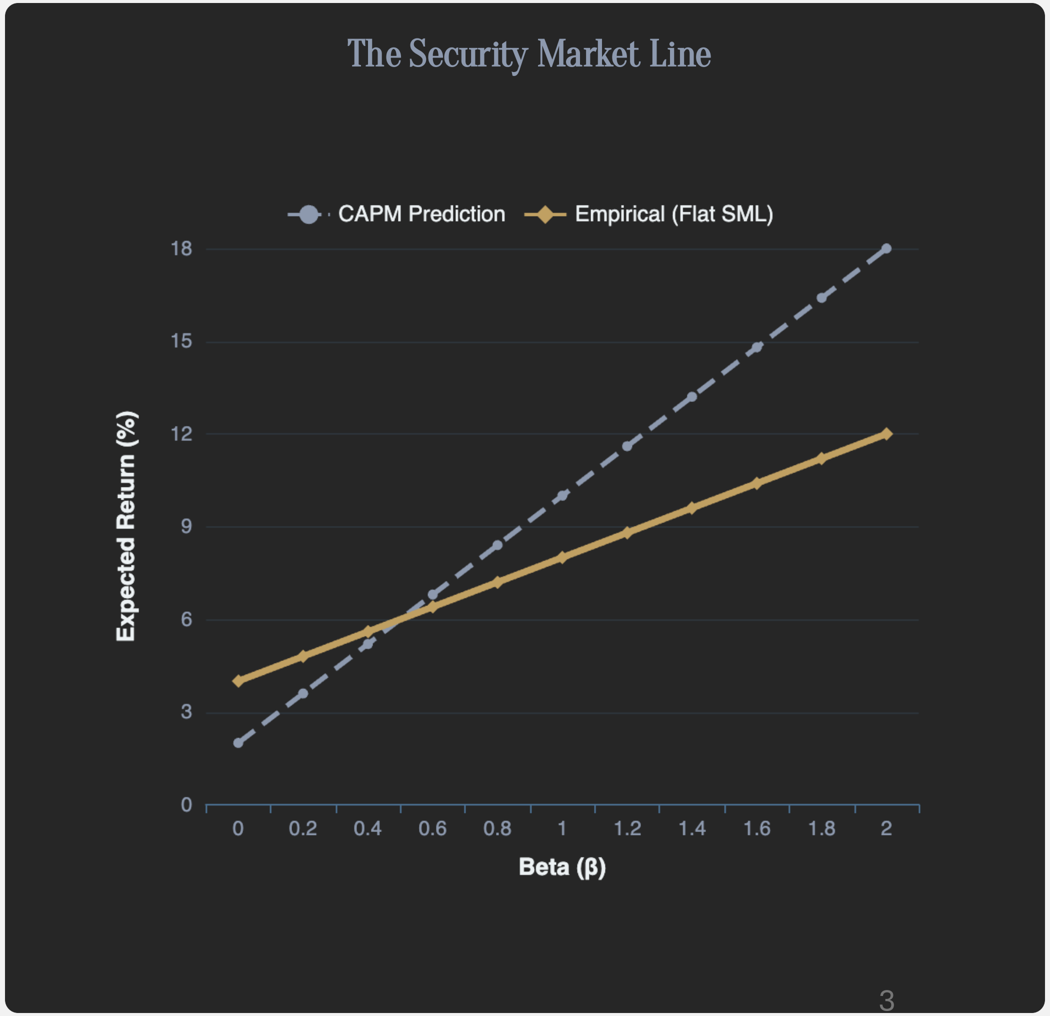
Introduction: Standard CAPM

CAPM Prediction: High-beta assets should earn proportionally higher returns. The Security Market Line (SML) should have a slope equal to the market risk premium

$$\text{CAPM: } E[R_i] = R_f + B_i * (E[R_i] - R_f)$$

- Y – intercept = Risk Free Rate (R_f)
- Slope = Beta(B_i)
- Market risk premium = ($E[R_i]$ - R_f)

- CAPM implies that higher risk (beta) assets lead to higher returns



Theory: Modified CAPM with Leverage

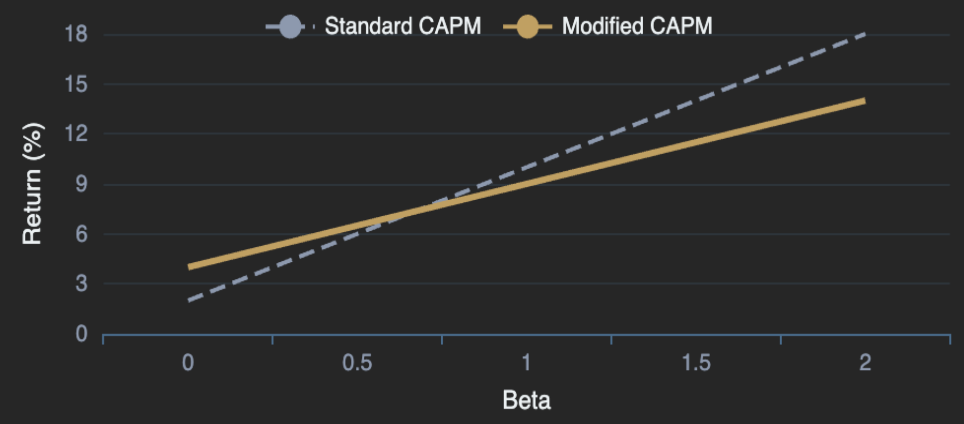
Theory: Given unconstrained leverage, the CAPM should be flatter since you can lever up on low beta assets

CAPM: $E[R_i] = R_f + B_i * \lambda + \psi$

- ψ = Lagrange Multiplier to measure funding constraint tightness
- Y – intercept = Risk Free Rate (R_f)
- Slope = Beta(B_i) * λ
- Market risk premium = λ
- $\lambda = E[R_m] - r_f - \psi$

- Tighter portfolio constraints (i.e., a larger ψ) flatten the security market line by increasing the intercept and decreasing the slope λ

Visual Comparison



Alpha and the Flat SML

$$\alpha_s = \psi(1 - \beta_s)$$

High β (> 1): Negative alpha ($\alpha < 0$)

Low β (< 1): Positive alpha ($\alpha > 0$)

β = 1: Zero alpha ($\alpha = 0$)

Data & Methodology

Time frame: US 1926 - 2012, Rest of world (19 countries) 1989 – 2012

January 1926 and March 2012

55,600 stocks, equities, bonds, FX, US treasury bonds, credit indices, corporate bonds, commodities, across 20 countries.

Data taken from: Center for Research in Security Prices (CRSP) tape

the Xpressfeed Globaldatabase

Trading signals: alphas with respect to the market factor and factor returns based on size (SMB), book-to-market (HML), momentum (up minus down, UMD), and(when available) liquidity risk

Beta: with respect to MSCI local index

Fama French model modified with leverage constraints

Standardised/normalised beta with the time series mean to increase significance

Data & Methodology

Country	Local market index	Number of stocks, total	Number of stocks, mean	Mean ME (firm, billion of US dollars)	Mean ME (market, billion of US dollars)	Start year	End year
Australia	MSCI Australia	3,047	894	0.57	501	1989	2012
Austria	MSCI Austria	211	81	0.75	59	1989	2012
Belgium	MSCI Belgium	425	138	1.79	240	1989	2012
Canada	MSCI Canada	5,703	1,180	0.89	520	1984	2012
Denmark	MSCI Denmark	413	146	0.83	119	1989	2012
Finland	MSCI Finland	293	109	1.39	143	1989	2012
France	MSCI France	1,815	589	2.12	1,222	1989	2012
Germany	MSCI Germany	2,165	724	2.48	1,785	1989	2012
Hong Kong	MSCI Hong Kong	1,793	674	1.22	799	1989	2012
Italy	MSCI Italy	610	224	2.12	470	1989	2012
Japan	MSCI Japan	5,009	2,907	1.19	3,488	1989	2012
Netherlands	MSCI Netherlands	413	168	3.33	557	1989	2012
New Zealand	MSCI New Zealand	318	97	0.87	81	1989	2012
Norway	MSCI Norway	661	164	0.76	121	1989	2012
Singapore	MSCI Singapore	1,058	375	0.63	240	1989	2012
Spain	MSCI Spain	376	138	3.00	398	1989	2012
Sweden	MSCI Sweden	1,060	264	1.30	334	1989	2012
Switzerland	MSCI Switzerland	566	210	3.06	633	1989	2012
United Kingdom	MSCI UK	6,126	1,766	1.22	2,243	1989	2012
United States	CRSP value-weighted index	23,538	3,182	0.99	3,215	1926	2012

Propositions Overview

High beta = low alpha because constrained investors look to high risk assets for returns

Betting against beta factor

--> Showed positive risk adjusted returns. BAB factor: long leveraged low-beta assets and short high-beta assets

When funding constraints tighten, BAB factor is low

More funding towards liquidity risk, pushed beta to 1 (market)

More constrained investors hold riskier assets

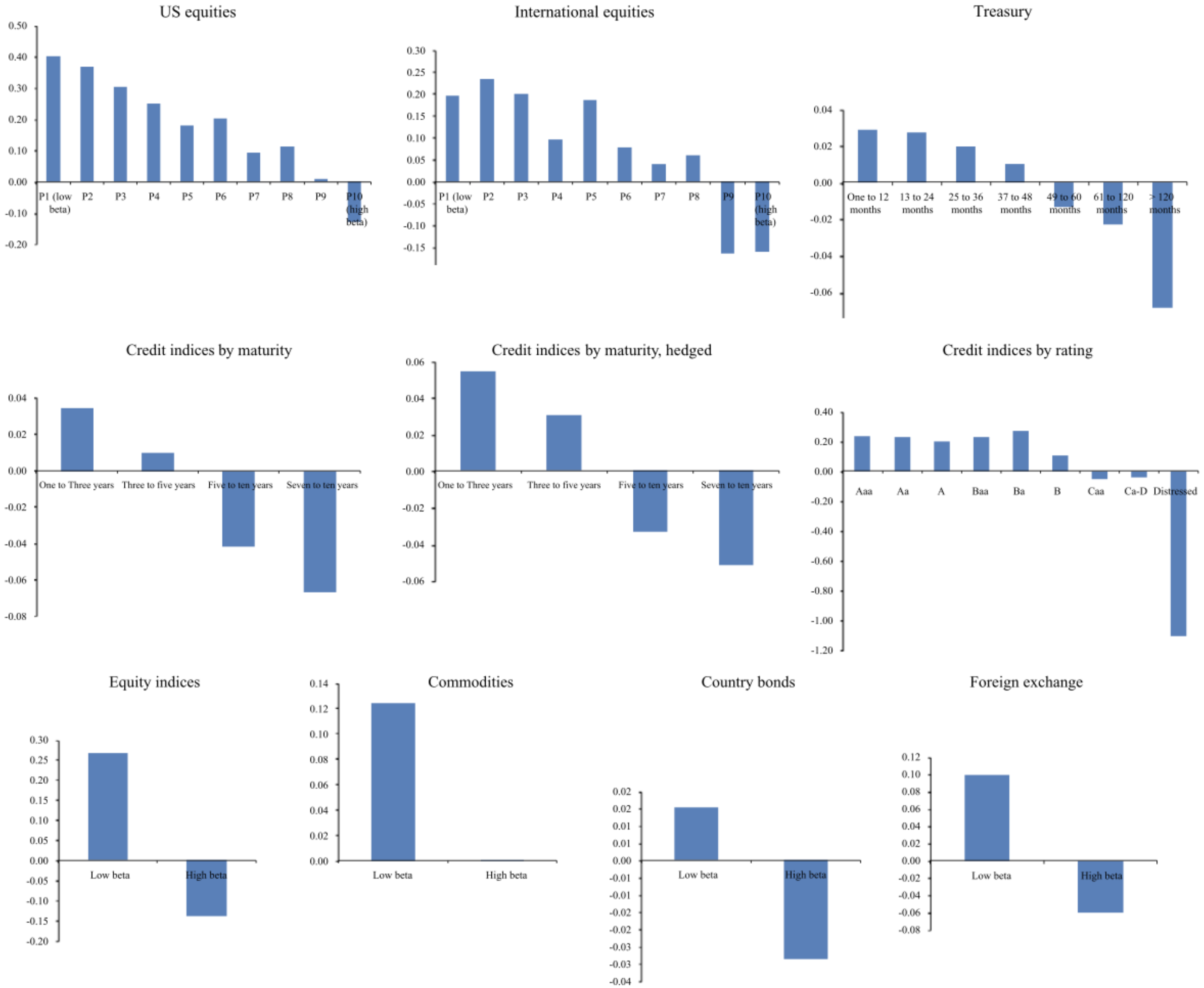
Proposition 1

Because constrained investors bid up high-beta assets, high beta is associated with low alpha

$$E_t(r_{t+1}^s) = r^f + \psi_t + \beta_t^s \lambda_t$$

It tests whether leverage constraints distort equilibrium pricing such that:

- High-beta assets are overpriced
- High-beta assets earn low risk-adjusted returns
- Low-beta assets earn positive alpha
- Empirically, this means testing → "*Does alpha decrease in beta?*"
- If yes → evidence that funding constraints affect asset prices.



- Consistent with Proposition 1 and with Black (1972), the alphas decline almost monotonically from the low-beta to high-beta portfolios.

Proposition 2

A betting against beta (BAB) factor, which is long leveraged low-beta assets and short high-beta assets, produces significant positive risk-adjusted returns.

$$E_t(r_{t+1}^{BAB}) = \frac{\beta_t^H - \beta_t^L}{\beta_t^L \beta_t^H} \psi_t \geq 0$$

It tests whether a market-neutral strategy that:

- Goes long leveraged low-beta assets
- Shorts high-beta assets earns a positive expected return due to funding constraints.
- Empirically, this means testing:
  - o Does the BAB factor generate positive abnormal returns?
  - o Are BAB returns stronger when funding constraints are tighter?

Portfolio	P1 (low beta)	P2	P3	P4	P5	P6	P7	P8	P9	P10 (high beta)	BAB
Excess return	**0.91**	**0.98**	**1.00**	**1.03**	**1.05**	**1.10**	**1.05**	**1.08**	**1.06**	**0.97**	**0.70**
	(6.37)	(5.73)	(5.16)	(4.88)	(4.49)	(4.37)	(3.84)	(3.74)	(3.27)	(2.55)	(7.12)
CAPM alpha	**0.52**	**0.48**	**0.42**	**0.39**	**0.34**	**0.34**	0.22	0.21	0.10	−0.10	**0.73**
	(6.30)	(5.99)	(4.91)	(4.43)	(3.51)	(3.20)	(1.94)	(1.72)	(0.67)	(−0.48)	(7.44)
Three-factor alpha	**0.40**	**0.35**	**0.26**	**0.21**	**0.13**	0.11	−0.03	−0.06	**−0.22**	**−0.49**	**0.73**
	(6.25)	(5.95)	(4.76)	(4.13)	(2.49)	(1.94)	(−0.59)	(−1.02)	(−2.81)	(−3.68)	(7.39)
Four-factor alpha	**0.40**	**0.37**	**0.30**	**0.25**	**0.18**	**0.20**	0.09	0.11	0.01	−0.13	**0.55**
	(6.05)	(6.13)	(5.36)	(4.92)	(3.27)	(3.63)	(1.63)	(1.94)	(0.12)	(−1.01)	(5.59)
Five-factor alpha	**0.37**	**0.37**	**0.33**	**0.30**	**0.17**	**0.20**	0.11	0.14	0.02	0.00	**0.55**
	(4.54)	(4.66)	(4.50)	(4.40)	(2.44)	(2.71)	(1.40)	(1.65)	(0.21)	(−0.01)	(4.09)
Beta (ex ante)	0.64	0.79	0.88	0.97	1.05	1.12	1.21	1.31	1.44	1.70	0.00
Beta (realized)	0.67	0.87	1.00	1.10	1.22	1.32	1.42	1.51	1.66	1.85	−0.06
Volatility	15.70	18.70	21.11	23.10	25.56	27.58	29.81	31.58	35.52	41.68	10.75
Sharpe ratio	0.70	0.63	0.57	0.54	0.49	0.48	0.42	0.41	0.36	0.28	0.78

- Consistent with Proposition 2, the BAB factor delivers a high average return and a high alpha.
- Shown is only US Equities, however, this result is consistent across all asset classes.

Proposition 3

When funding constraints tighten, the return of the BAB factor is low

$$\frac{\partial r_t^{BAB}}{\partial m_t^k} \leq 0 \qquad \frac{\partial E_t(r_{t+1}^{BAB})}{\partial m_t^k} \geq 0$$

It tests whether BAB is exposed to funding liquidity risk.

- Does BAB perform poorly when funding conditions tighten?
- Do funding shocks compress betas toward 1?
- Does BAB earn higher subsequent returns after funding stress?

If yes → BAB is compensation for funding liquidity risk.

Left-hand side: BAB return	US equities (1)	(2)	International equities, pooled (3)	(4)	All assets, pooled (5)	(6)
Lagged TED spread	**−0.025**	**−0.038**	**−0.009**	**−0.015**	**−0.013**	**−0.018**
	(−5.24)	(−4.78)	(−3.87)	(−4.07)	(−4.87)	(−4.65)
Change in TED spread	**−0.019**	**−0.035**	**−0.006**	**−0.010**	**−0.007**	**−0.011**
	(−2.58)	(−4.28)	(−2.24)	(−2.73)	(−2.42)	(−2.64)
Beta spread		0.011		0.001		0.001
		(0.76)		(0.40)		(0.69)
Lagged BAB return		0.011		0.035		0.044
		(0.13)		(1.10)		(1.40)
Lagged inflation		−0.177		0.003		−0.062
		(−0.87)		(0.03)		(−0.58)
Short volatility return		**−0.238**		0.021		0.027
		(−2.27)		(0.44)		(0.48)
Market return		**−0.372**		**−0.104**		**−0.097**
		(−4.40)		(−2.27)		(−2.18)
Asset fixed effects	No	No	Yes	Yes	Yes	Yes
Number of observations	328	328	5,725	5,725	8,120	8,120
Adjusted R^2	0.070	0.214	0.007	0.027	0.014	0.036

- Consistent with Proposition 3, BAB returns decline when funding conditions tighten and increase following funding stress.
- TED Spread is the difference between short-term U.S. government debt (T-bills) and interbank loans (LIBOR)
- We use TED Spread as a proxy of funding conditions

Proposition 4

Increased funding liquidity risk compresses betas toward one

- Example of funding volatility:
  - Higher volatility in the TED spread
  - Periods of credit stress
  - Financial crisis environments
- Implication:
  - Cross-sectional beta dispersion declines, and BAB temporarily acquires positive market beta

	Alpha	Conditional market beta			
		P1 (low TED volatility)	P2	P3 (high TED volatility)	P3 – P1
Panel D: US equities					
CAPM	**1.06**	**– 0.46**	– 0.19	– 0.01	**0.45**
	(3.61)	(– 2.65)	(– 1.29)	(– 0.11)	(3.01)
Control for three factors	**0.86**	**– 0.40**	– 0.02	0.08	**0.49**
	(4.13)	(– 3.95)	(– 0.19)	(0.69)	(3.06)
Control for four factors	**0.66**	**– 0.28**	0.00	0.13	**0.40**
	(3.14)	(– 5.95)	(0.02)	(1.46)	(4.56)
Panel E: International equities					
CAPM	**0.60**	– 0.09	0.02	0.06	0.16
	(2.84)	(– 1.30)	(0.64)	(1.28)	(1.87)
Control for three factors	**0.59**	– 0.09	0.02	0.05	0.14
	(3.23)	(– 1.22)	(0.74)	(1.09)	(1.70)
Control for four factors	**0.35**	– 0.04	0.05	**0.07**	**0.11**
	(2.16)	(– 1.16)	(1.51)	(2.03)	(2.24)
Panel F: All assets					
CAPM	**0.54**	**– 0.13**	– 0.07	0.01	**0.14**
	(4.96)	(– 2.64)	(– 1.82)	(0.21)	(2.34)

- Consistent with Proposition 4, BAB's conditional market beta increases significantly when high TED volatility is high, indicating beta compression during funding stress.
- Demonstrates that the BAB factor fails to stay market neutral when it matters most.

Proposition 5

More constrained investors hold riskier assets.

- *Example of a constrained investor:*
  - *Mutual Funds because of restrictions on leverage (1940 Investment Company Act)*
  - *Retail Investors*
- *Example of an unconstrained investor:*
  - *Private Equity firms (Especially those attempting leverage buyouts)*
  - *Berkshire Hathaway*

Investor, method	Sample period	Ex ante beta of positions		Realized beta of positions	
		Beta	*t*-Statistics (H0: beta=1)	Beta	*t*-Statistics (H0: beta=1)
Panel A: Investors likely to be constrained					
Mutual funds, value weighted	1980–2012	**1.08**	2.16	**1.08**	6.44
Mutual funds, equal weighted	1980–2012	1.06	1.84	**1.12**	3.29
Individual investors, value weighted	1991–1996	**1.25**	8.16	**1.09**	3.70
Individual investors, equal weighted	1991–1996	**1.25**	7.22	**1.08**	2.13
Panel B: Investors who use leverage					
Private equity (all)	1963–2012	0.96	−1.50		
Private equity (all), equal weighted	1963–2012	**0.94**	−2.30		
Private equity (LBO, MBO), value weighted	1963–2012	**0.83**	−3.15		
Private equity (LBO, MBO), equal weighted	1963–2012	**0.82**	−3.47		
Berkshire Hathaway, value weighted	1980–2012	**0.91**	−2.42	**0.77**	−3.65
Berkshire Hathaway, equal weighted	1980–2012	**0.90**	−3.81	**0.83**	−2.44

- Consistent with Proposition 5 in that investors with less leverage constraints tend to acquire assets with low betas.

Conclusion: Constraint on Arbitrage

Constraint:
Borrowing limits force investors to bid up high-beta assets

Behavior:
Proposition 5: Mutual funds reach for yield; smart money leverages safety.

Result:
Security Market Line flattens globally

Opportunity:
BAB captures this premium across Equities, Bonds and Credit